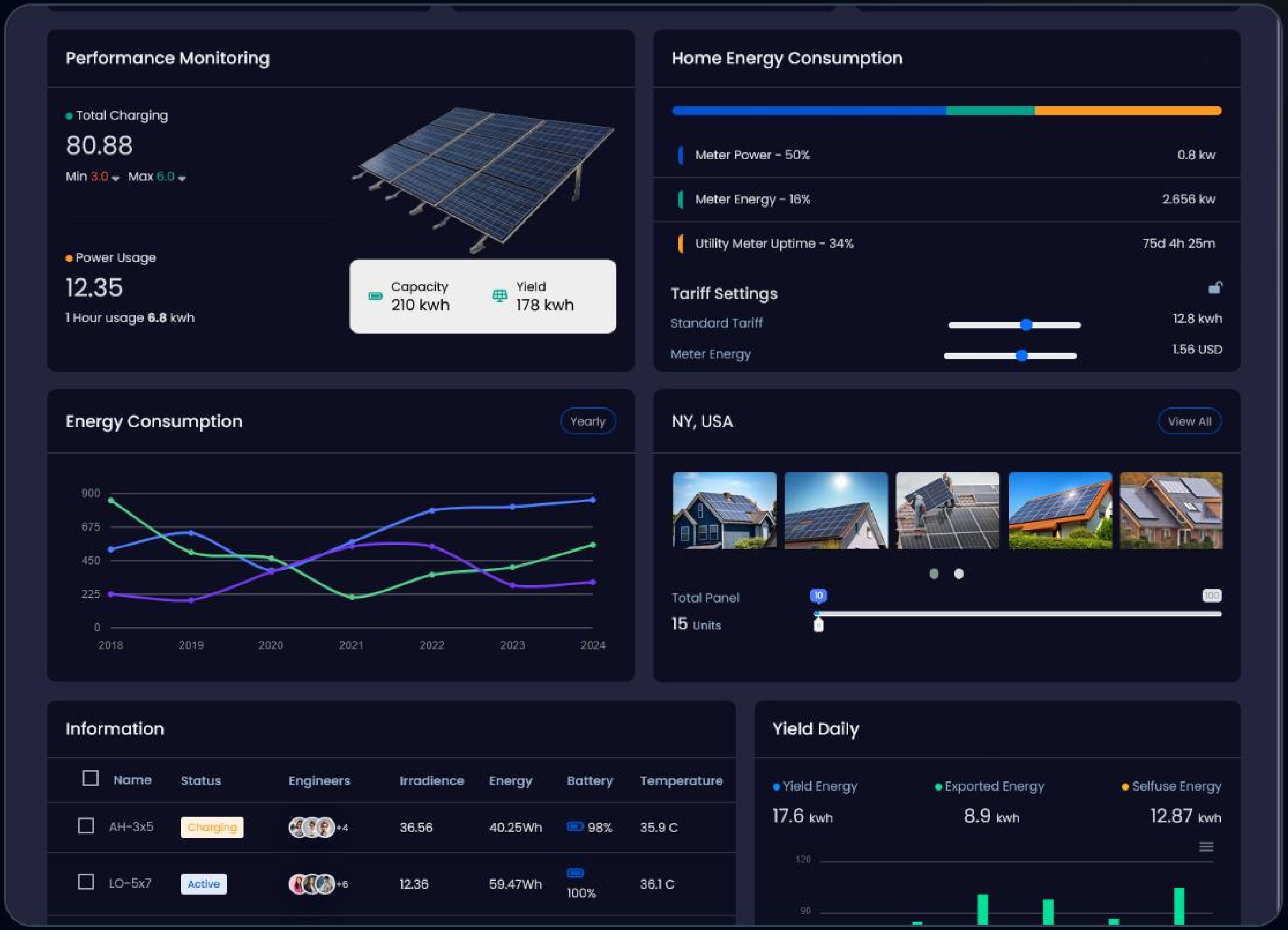
Solar Load Calculation

The Engineering Blueprint for Sizing Residential Systems and Battery Runtime



Categorizing Consumption

Understanding the difference between what you use and what you need.

Load Categorization

Continuous Loads

Appliances that draw a steady, predictable amount of power over time, such as LED lighting, TVs, and Internet routers. Measured in Watt-Hours (Wh).

Inductive Loads

Appliances with mechanical motors like pumps and AC compressors. These require a massive "Surge" spike just to start moving stationary rotors.

Common Appliance Wattages

Appliance	Continuous (Watts)	Surge Requirement	Load Type
LED Lighting (x10)	100 W	None	Resistive
Inverter Refrigerator	200 W	600 W	Inductive
1.5HP Inverter AC	1,050 W	1,200 W	Smart Inductive
Water Pump (1HP)	750 W	3,750 W	Heavy Inductive

Surge Demand Comparison

High surge factors (3x-5x) dictate the minimum size of your inverter, even if your running load is low.

The "Sumo" Surge Factor

5X

Peak Startup Spike

The Inductive Rush

For a 1HP water pump drawing 750W continuously, your inverter must be capable of handling a **3,750W surge** for 1–2 seconds to overcome mechanical friction.

Battery Storage Mathematics

Calculating how long your power will actually last.

The 80% Safety Boundary

Draining Lithium past 80% DoD rapidly accelerates chemical degradation and reduces lifespan by years.

Calculating Usable Energy

- **The Core Capacity:** Voltage (V) multiplied by Amp-Hours (Ah) gives total Watt-Hours (Wh).
- **Applying the Buffer:** Multiply total capacity by 0.8 to find your "Usable" energy window.
- **Runtime Result:** Divide Usable Wh by your continuous Load (W) to find total hours.

$$\text{Runtime (Hours)} = \frac{(\text{V} \times \text{Ah}) \times 0.8}{\text{Continuous Load (W)}}$$

The 1.5HP AC Case Study

Configuration: 48V 100Ah LiFePO4 Battery

Total Wh: 4,800Wh

Usable Wh (80%): 3,840Wh

AC Draw: 1,050W Continuous

Runtime: 3.6 Hours Total

Conclusion: 100Ah is insufficient for all-night AC cooling.

The Golden Window Strategy

10 AM

Solar harvest begins. Run lighter pumps.

12 PM

Peak Harvest. Run heavy water pumps (Sumo).

3 PM

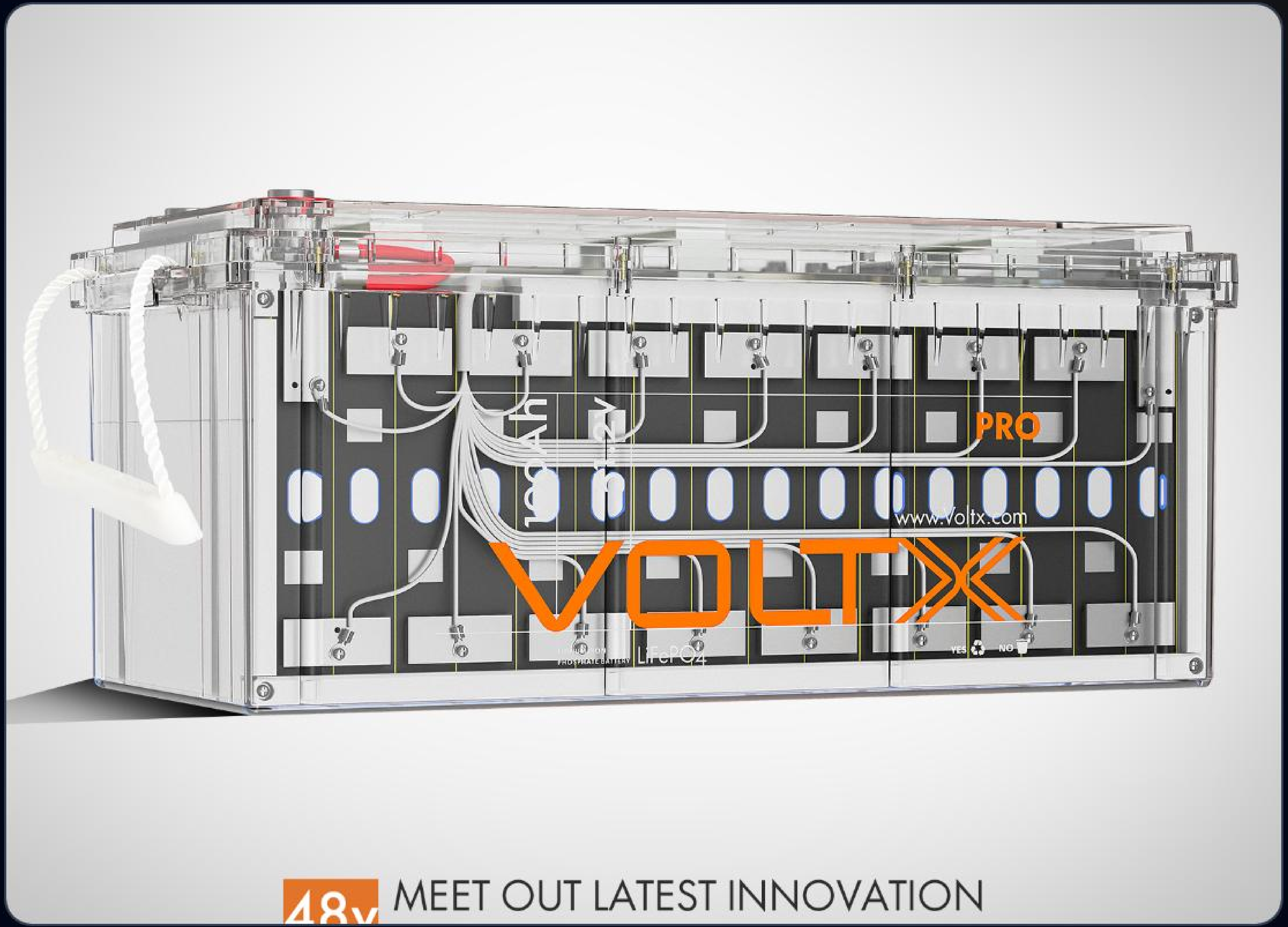
Recharge buffer. High production ends.

8 PM

Battery discharge. Only priority loads.

Ready to Automate?

Next: Behind-The-Scenes Tech & AI Operations



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